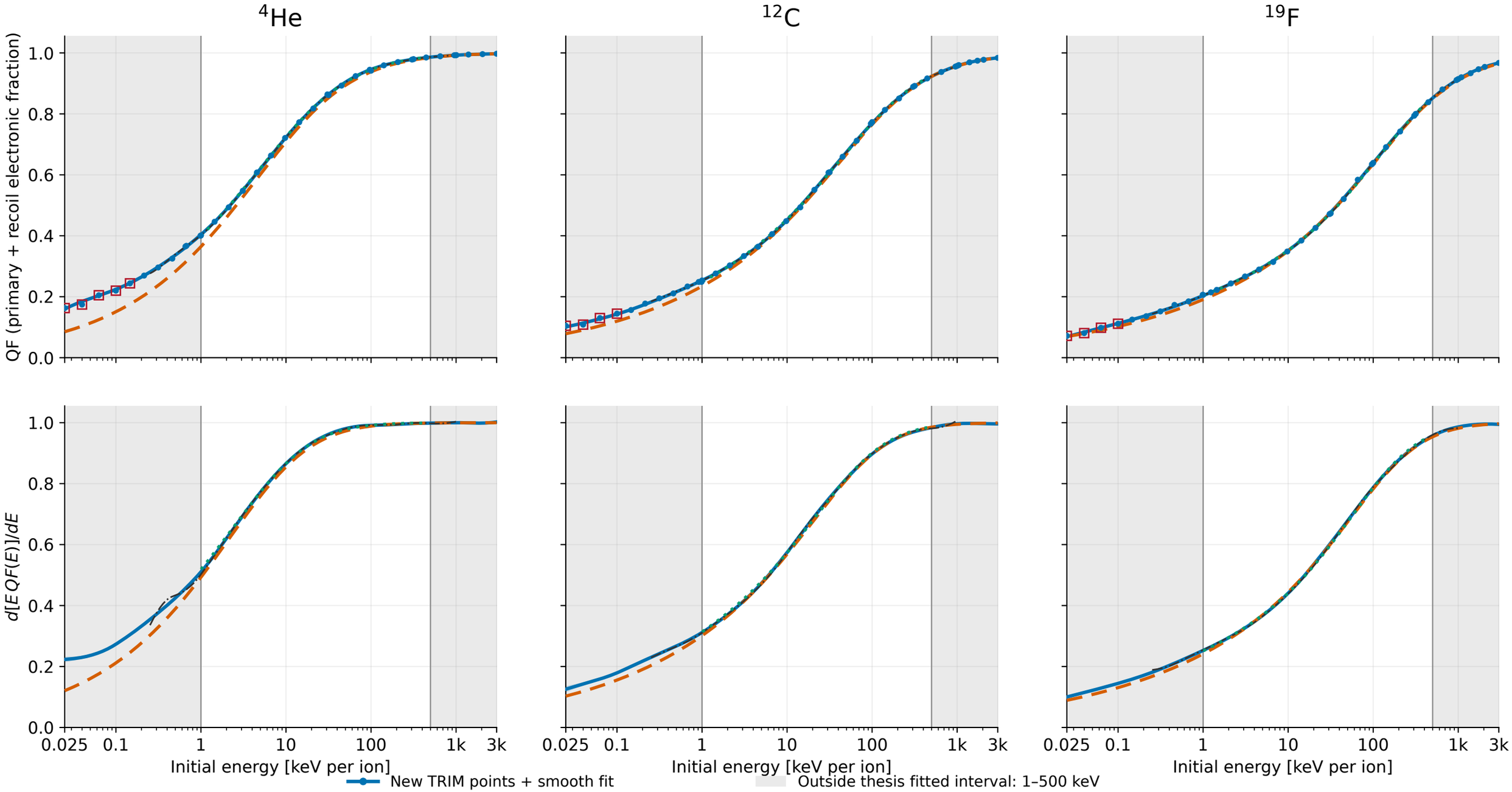
CYGNO quenching recap — thesis implementation vs completed TRIM scan

He/CF_4 60/40 molar · new direct simulations: 900 mbar, 293 K · electronic-energy response

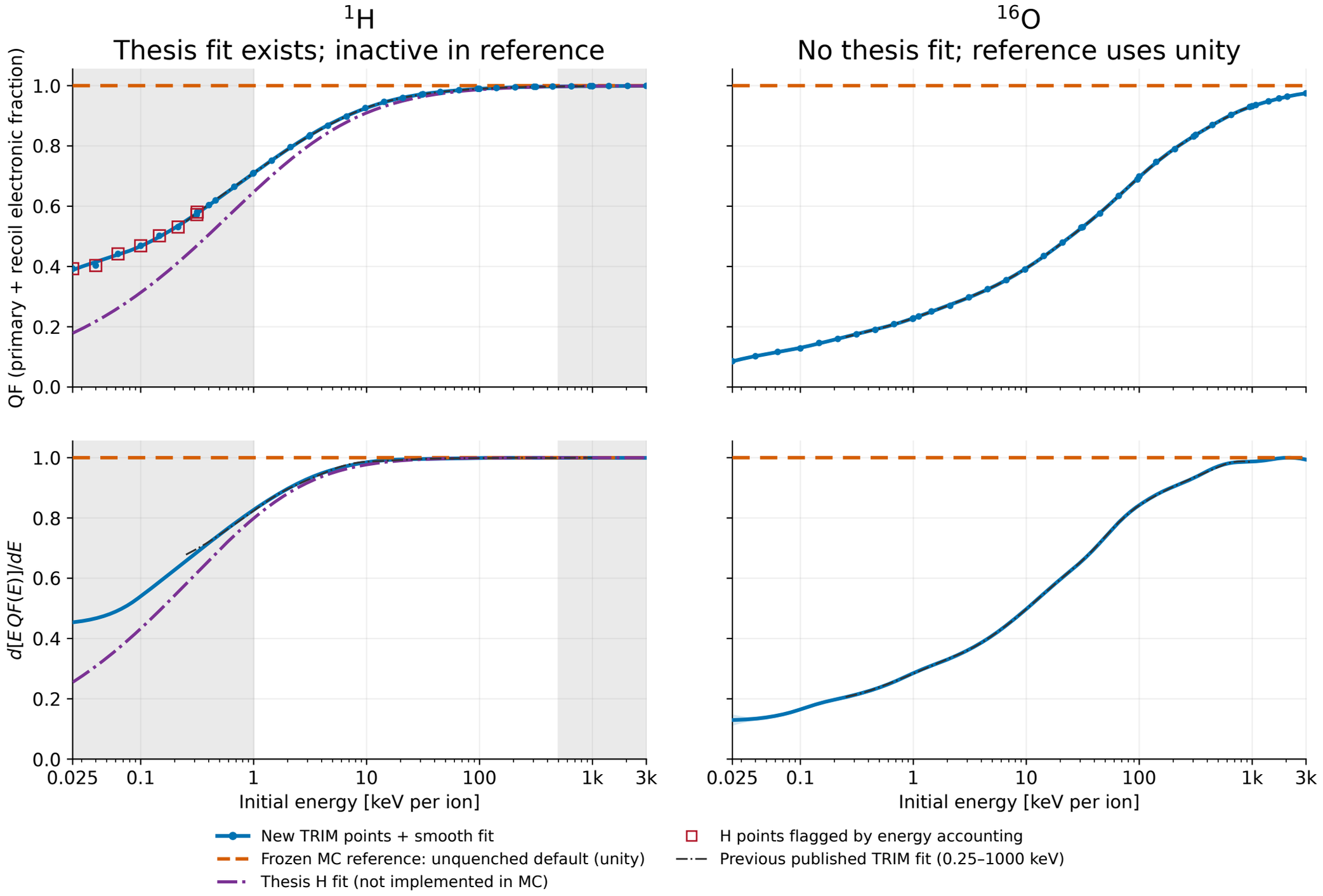


Blue bands: ±1 conditional statistical SEM; systematic/model uncertainty excluded. Green refit is not a validated replacement.

16 Oct 2026 MC reference: He/C/F fits; H/O unity defaults. The thesis supplies an H fit, but no O fit.

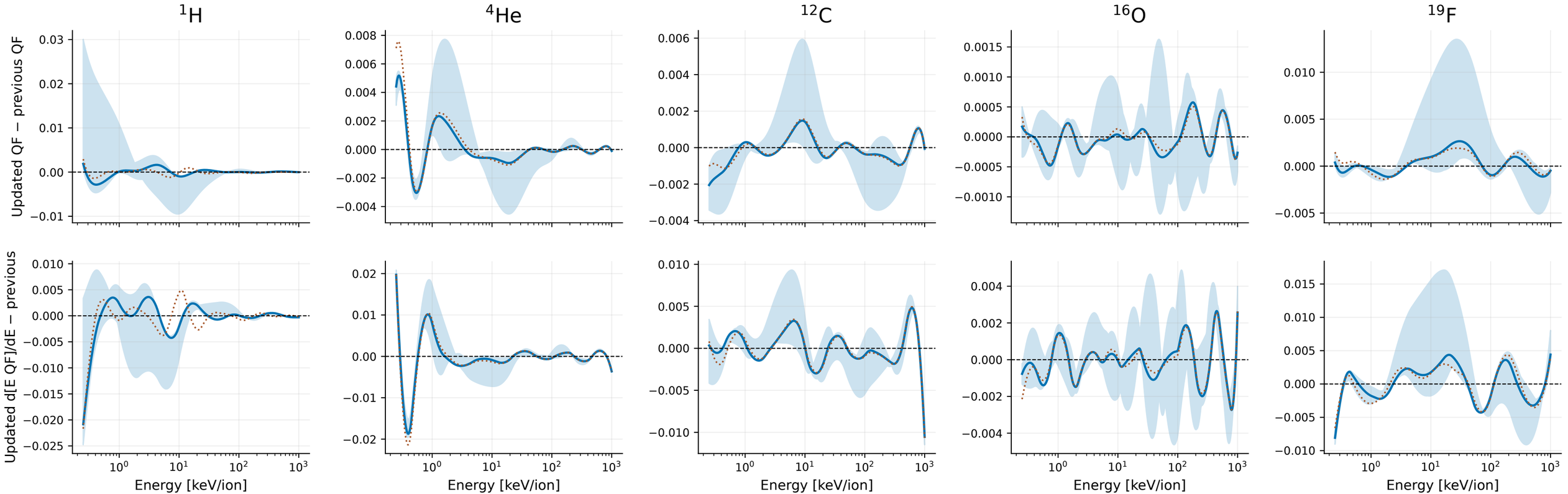
CYGNO recap — new hydrogen and oxygen recoil responses

He/CF_4 60/40 molar · 900 mbar, 293 K · full-cascade TRIM



Blue bands: ±1 conditional statistical SEM; model/calibration uncertainty excluded. Red squares retain closure warnings.

Grey applies to the thesis H fit only: extrapolation outside 1–500 keV. Unity lines are code defaults, not physical QF predictions.


Effect of extending the fit — comparison over the previous 0.25–1000 keV interval
^{1}H
^{4}He
^{12}C
^{16}O
^{19}F
Updated QF − previous QF
Updated d[E QF]/dE − previous
Energy [keV/ion]
Blue: updated fit minus previous published fit. Shading: basis/penalty sensitivity envelope; NOT a statistical confidence interval.
Brown dotted: extended fit excluding the new 25–63 eV anchors. Fits share data; change uncertainties are unassigned.

CYGNO QF — thesis cross-check

CYGNO QF extension — 17 October 2026

What was added and reused
All 45 bounded full-cascade batches completed: H1, He4, C12, O16 and F19 at 25, 40 and 63 eV, three independent 100-history batches per point. No adaptive campaign or additional systematic scan was launched. All 852 compatible prior batches were reused, including direct points below 250 eV and above 1 MeV. The combined 195 direct points support 0.025–3000 keV per ion. Compatible nominal coarse controls retain their separate source tags; threshold variations, pilots and correlated fine controls are not pooled.

Settings and accounting
The direct gas remains He/CF4=60/40 molar, 900 mbar, 293 K, rho=0.001389194959 g/cm3, He:C:F atomic fractions=3:2:8. TRIM mode 5, gas flag 1, compound correction 1 and displacement 21 eV were unchanged. The requested 293.15 K target still uses an approximate transfer, not direct target simulations.
Primary plus recoil IONIZ energy is counted once. Every new archive member hash and independent electronic/phonon integration passed. New maximum absolute closure residual=14.582%; 45 new batches exceed the retained 1% warning criterion.

Statistics and model limits
Only 6 of 15 new points meet relative QF SEM<=2%; all remain at the requested 300 histories. There is no automatic expansion. These energies approach the 21 eV displacement setting; 25 eV is only 1.19 times that value. That setting is not an ionization threshold, and its physical adequacy is unvalidated here. Existing 1 keV cutoff controls do not validate 25–63 eV physics. Electronic energy includes excitation and is not measured charge.

Extended smooth response and change from previous release The same bounded monotone cubic B-spline in log(E) now fits the full measured support. Basis size (12/14/16/18/20 coefficients) and regularization are selected jointly by leave-one-energy-out cross-validation; 400 joint-channel batch bootstraps give conditional statistical SEMs. d[E QF]/dE is analytic from that same model. Measured endpoint data support the fitted values; the evaluator rejects inputs outside the explicit bounds. The previous fit was not held fixed. Over 0.25–1000 keV: H1: max |delta QF|=0.00285; max |delta dR/dE|=0.02083. He4: max |delta QF|=0.00518; max |delta dR/dE|=0.01972. C12: max |delta QF|=0.00205; max |delta dR/dE|=0.01054. O16: max |delta QF|=0.00058; max |delta dR/dE|=0.00274. F19: max |delta QF|=0.00266; max |delta dR/dE|=0.00807.

Derivative qualification
Refits with two fewer/more coefficients and weaker/stronger regularization quantify basis/penalty sensitivity. Their envelope is not a confidence interval and is not folded into statistical SEM.
Endpoint derivatives are one-sided model values; global derivative stability is not established. The change plot shows the difference from the prior response and this sensitivity on its old interval. Both fits reuse correlated data, so change uncertainties remain unassigned.

Thesis and MC interpretation
Flaminia Di Giambattista's thesis, Sec. 4.1.2.1, Table 4.1 and Eq. 4.17, uses the same inclusive IONIZ definition and response derivative. Its fitted domain is 1–500 keV; shaded curves beyond that domain extrapolate. He/C/F coefficients match frozen MC. The thesis H fit is not active in the reference; O has no thesis fit. Orange H/O unity lines are code defaults, not physical predictions. Per-hit use of inclusive cascades needs ancestry/terminal-energy bookkeeping. See REPORT.md and the machine-readable validation for integration limits.